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Winch Equations

27 March 2008

Introduction

For successful operation, our wind sensor needs a winch calibration that relates deployed tether length to the number of winch revolutions as measured by a bicycle odometer. Because the winch radius decreases as tether is deployed, the desired relationship is not linear.

Nomenclature

Symbol	Definition
d	Tether line diameter
w	Winch drum width
l	Length of deployed tether
r	Winch drum radius
N	Number of winch revolutions since tether deployment began

Analysis

The objective is to develop a relation of the form, $\mathbf{l = f(N)}$. First, assume that the line is wound uniformly and evenly over the drum. Even though this is perhaps a poor assumption, it will still provide an indication of what to expect. Then, the number of turns, **n**, to cover the drum width with one layer of tether is

$$\mathbf{n = w/d}.$$

The change in **l** as one drum-width of tether is deployed is

$$\mathbf{\Delta l = 2 \pi r w / d}.$$

Now,

$$\mathbf{\Delta r/\Delta l \approx dr/dl = -d/(2 \pi r w / d) = -d^2 / 2 \pi r w}, \text{ or}$$

$$\mathbf{r\, dr = -(d^2 / 2 \pi w)\, dl}, \text{ or}$$

$$\mathbf{r^2 - ro^2 = -d^2 l / 2 \pi w},$$

where **ro** is the drum radius before any tether is deployed. Then,

$$\mathbf{r = \sqrt{ro^2 - d^2 l / 2 \pi w}}.$$

Next,

$$\mathbf{\Delta l/\Delta N = dl/dN = 2 \pi r = 2 \pi \sqrt{ro^2 - d^2 l / 2 \pi w}}, \text{ or}$$

$$\mathbf{dl} \,/\, \sqrt{\mathbf{ro}^2 - \mathbf{d}^2\,\mathbf{l} / 2\,\pi\,\mathbf{w}} = 2\,\pi\,\mathbf{dN}, \text{ or}$$

$$\mathbf{N} = (2\,\mathbf{w} / \mathbf{d}^2)\left[\mathbf{ro} - \sqrt{\mathbf{ro}^2 - \mathbf{d}^2\,\mathbf{l} / 2\,\pi\,\mathbf{w}}\right].$$

This can easily be inverted to give **l** as a function of **N**:

$$\mathbf{l} = (2\,\pi\,\mathbf{w} / \mathbf{d}^2)\left[\mathbf{ro}^2 - (\mathbf{d}^2\,\mathbf{N} / 2\,\mathbf{w} - \mathbf{ro})^2\right], \text{ or}$$

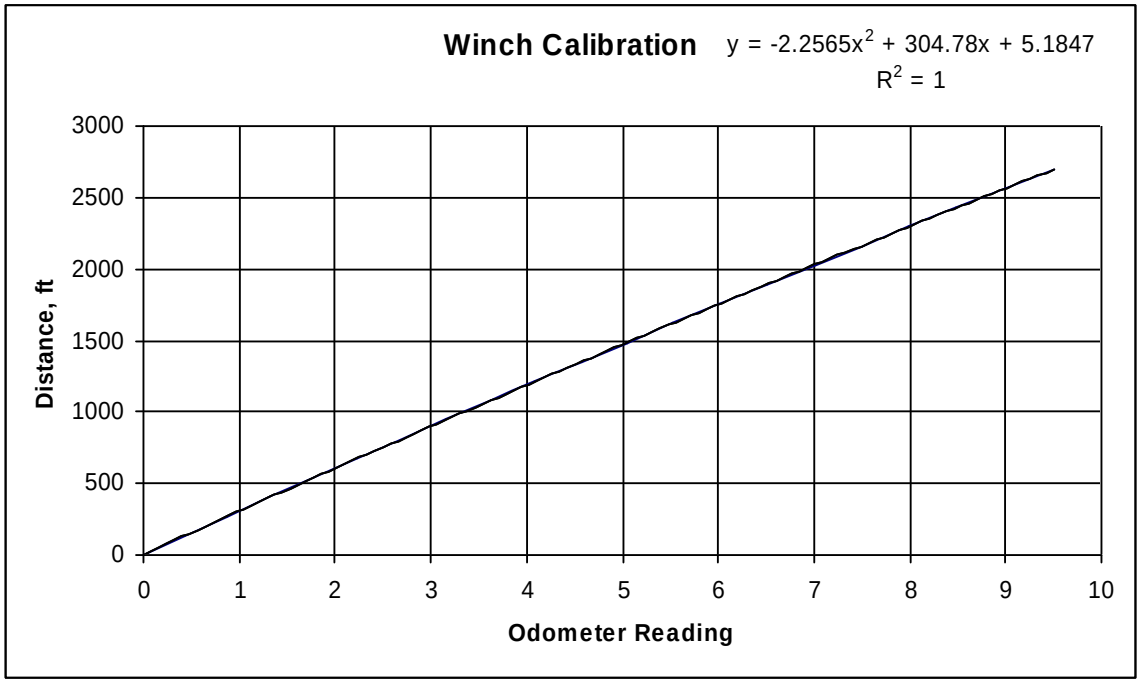
$$\mathbf{l} = 2\,\pi\,(\mathbf{ro}\,\mathbf{N} - \mathbf{d}^2\,\mathbf{N} / 4\,\mathbf{w}).$$

This is the result that should guide the formula used in the Launch Day Tool. The first term is just what we'd expect if the drum radius did not shrink as tether line is deployed. The second term is a quadratic correction accounting for the shrinking drum radius as tether is deployed.

To implement this in the LDT, it's necessary to have an equation of the form,

$$\mathbf{l} = \mathbf{AN} + \mathbf{BN}^2.$$

Given some calibration data, plot the data in Excel, and fit a quadratic line to this as shown in the figure below:



This is the winch calibration curve for our operational sensor.
Q.E.D.